# LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

## Quick Revision: Work and Energy

### Key terms

| Quantity | Formula | SI unit |
|---|---|---|
| Work | W = F × s (force × displacement in its direction) | joule (J) |
| Kinetic energy | KE = ½mv$^2$ | J |
| Potential energy | PE = mgh | J |
| Power | P = W/t | watt (W) |
| Commercial unit of energy | 1 kWh = 3.6 × 10$^6$ J | kWh ('unit') |

### Must-know points

- Work is done only if a force moves the body. **No displacement means no work** (e.g. pushing a wall).
- Work is **positive** when force and displacement are in the same direction, **negative** when opposite (e.g. friction), and **zero** when they are at right angles.
- 1 J of work is done when a force of 1 N moves a body 1 m in its direction.
- **Law of conservation of energy:** energy can only change from one form to another; the total stays constant.
- A freely falling body: PE changes into KE, and PE + KE stays the same at every point.
- Doubling the speed makes KE **four times**, because KE depends on v$^2$.

Take g = 9.8 m/s$^2$ unless the question says 10 m/s$^2$.
1 kW = 1000 W · 1 hour = 3600 s · Energy (kWh) = power (kW) × time (h)

### Common mistakes

- Using mass in grams. Convert to kg first.
- Forgetting to square the velocity in ½mv$^2$.

### Quick check

1. Find the potential energy of a 2 kg ball held 5 m above the ground (g = 10 m/s$^2$).

2. A 60 W bulb glows for 10 hours. How many units of energy does it use?